

1. Load a dataset of sales transactions. Find the total sales made per region.
2. Given student attendance data, find students with attendance < 75%.
3. Analyze a COVID-19 dataset: show total confirmed cases per state.
4. From a weather dataset, find the hottest day for each city.
5. Find the correlation between two numeric columns in a dataset.
6. Merge two DataFrames: one with student details, another with their marks.
7. Create a bar plot of average marks scored per department.
8. Analyze a file with exam results. Find the top 3 scorers per subject.
9. From a dataset of employee records, filter employees who joined before 2020.
10. Generate descriptive statistics grouped by gender for a health dataset.
11. Group the employee data by department and compute the average salary.
12. Find the product with the highest revenue (price × quantity).
13. Add a pass/fail column to marks data (pass if marks ≥ 40).
14. Find the highest electricity consumption day for each month.
15. Extract the hour from a timestamp and count how many transactions occurred each hour.
16. Calculate total revenue per customer and rank them.
17. Plot average monthly rainfall for each city.
18. Use groupby to find the maximum marks scored in each subject.
19. Find the most watched movie by total number of ratings.
20. Calculate the win percentage for each team.
21. Calculate daily returns from stock price data.
22. Compute percentile rank for each student.
23. Use pivot\_table to summarize sales data by region and month.
24. Identify missing months and fill them with 0.
25. Average steps walked per weekday from fitness tracker.

26. Count the number of patients per disease category.
27. Find top 5 most accident-prone roads from traffic data.
28. Compute number of days exceeding 40°C per year.
29. Group and count feedbacks by rating.
30. Count number of posts made per day and plot trend.
31. Blood bank- units donated per day.
32. Reservation counters-visualize daily bookings.

**Note: with above questions which are from unit 4, anyone question from Units – 1, 2 and 3 will be given in internal as well as external lab exam. (2 questions for every student)**

- Program to implement Newton-Raphson method.
- Program to implement Bisection method.
- Program to implement Trapezoidal Rule.
- Stack and Queue implementation.
- Searching and Sorting programs.
- List, Tuple, Sets and Dictionaries related programs.
- Functions and Loop examples.